

App Assumptions & Estimations (PROD)

About the Product & Use Cases

The application is a **carbon footprint registry and certification system**, designed for **country-level adoption** and long-term operation.

The PROD environment represents the **live, externally facing system**, used by multiple organizations and end users.

Primary use cases include:

- Document upload and management
- Report generation
- Status tracking of submissions and certifications
- User notifications and communications

System characteristics:

- No external business-logic integrations
- No batch operations, except:
 - Internal maintenance tasks
 - Operational housekeeping

The backend architecture is **stateless**, supporting horizontal scalability and predictable performance.

About Traffic & Load

Expected production traffic characteristics:

- **200 Daily Active Users (DAU)**
- **~1,000 API requests per user per day**

Estimated request volume:

- **~200,000 API requests per day**
- **~6,000,000 API requests per month**

Peak traffic:

- **~20 requests per second (RPS)**

Traffic pattern:

- Concentrated during working hours
- Evenly distributed across ~12 hours/day
- No bursty or bulk-user behavior
- No large batch operations triggered by users

The backend remains stateless and horizontally scalable.

About the API and Containers

API characteristics:

- Same codebase across all environments
- Stateless design
- Optimized for request/response workflows

Performance assumptions:

- **~2 seconds execution time per request (worst-case)**
- Typical execution time is lower under normal conditions

These values are used for:

- Autoscaling calculations
 - Concurrency estimation
 - Cost modeling
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About the Database

Database characteristics:

- Relational database (PostgreSQL)
- No PostGIS or geospatial extensions
- Simple SQL queries (CRUD + joins)
- No heavy analytics workloads

Schema characteristics:

- ~45 tables
- All tables use foreign keys

Growth assumptions:

- Database size is expected to **double every year**
- No additional IOPS are explicitly provisioned
- Storage scales independently as needed

The database is designed for transactional consistency rather than analytical workloads.

About the Storage

The storage layer is used to store:

- User-uploaded documents
- Images
- Generated PDF reports
- Application metadata

Usage characteristics:

- Uploads are user-driven
- No bulk ingestion pipelines
- No streaming or large-scale file imports

Growth expectations:

- Data accumulates across multiple organizations

- Expected growth from **hundreds of GB to TB-scale over time**
- Long-term retention is required

Storage is treated as a durable, production-grade repository.

About Logs & Observability

Observability is provided through Azure Monitor and Application Insights.

Logging assumptions:

- Application logs ingested with **~10% sampling enabled**
- Focus on:
 - Errors
 - Performance insights
 - Operational diagnostics

Retention:

- **3 months (90 days)**

This provides sufficient data for:

- Debugging
 - Incident analysis
 - Operational monitoring
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About Authentication

Authentication is handled using **Microsoft Entra ID**.

User assumptions:

- Current active usage aligned with DAU assumptions
- **Expectation of growth up to ~10,000 Monthly Active Users (MAU)** over time

The authentication model is designed to support:

- Multi-organization access

- Future scale without architectural change
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About Email / Notifications

Emails are used for:

- Account-related notifications
- Status updates
- Workflow and submission events

Email characteristics:

- **Average email size:** ~5 MB
- Transactional usage only
- No marketing or bulk campaigns

Delivery volume scales with user growth but remains predictable.

About Reliability & Failure Tolerance

The PROD environment targets production-grade reliability appropriate for a national-level regulatory platform:

- **Acceptable error rate:** $\leq 0.5\%$ of requests (server-side errors)
- **API timeout threshold:** 10 seconds (hard timeout)
- **Recovery Time Objective (RTO):** ≤ 4 hours
- **Recovery Point Objective (RPO):** ≤ 15 minutes

These targets reflect the non-real-time nature of the system while ensuring strong data integrity, user trust, and operational credibility.

Acceptable Error Rate Explanation

The recommendation is $\leq 0.5\%$ of requests resulting in server-side errors (5xx)

This makes sense because:

- Users are performing **important but non-real-time operations**
- Occasional retries are acceptable
- Errors should be:
 - Rare
 - Visible
 - Recoverable

Interpretation:

- 99.5% success rate minimum
- Errors should not:
 - Corrupt data
 - Lose submissions
 - Block certification flows

Comparison:

System Type	Typical Error Rate
Consumer SaaS	$\leq 1\%$
Regulatory / gov-adjacent	$\leq 0.5\%$
Financial trading	$\leq 0.1\%$

✓ **0.5% is a strong but realistic target**

Timeout Threshold Explanation

The recommendation is 10 seconds hard timeout for API requests

This makes sense because:

- Most operations:
 - Document upload
 - Validation

- Status updates
- Are **not latency-critical**
- AI-assisted workflows may legitimately take longer

Here is a practical breakdown:

- Normal requests: 1–3 seconds
- Slow requests: 3–6 seconds
- Hard cutoff: 10 seconds

Anything beyond this should:

- Fail cleanly
- Be retried
- Or move to async processing

Comparison:

System Type	Timeout
Real-time trading	<1s
E-commerce	3–5s
Regulatory platforms	8–12s

✓ **10s is well-aligned**

RTO – Recovery Time Objective Explanation

The recommendation is ≤ 4 hours

This mean if PROD goes down:

- The system must be **operational again within 4 hours**

This is appropriate because:

- No life-critical or financial market impact
- Users operate during business hours
- Temporary outages can be communicated

Comparison:

Platform Type	RTO
Mission-critical banking	<1 hour
National registries / compliance systems	2–8 hours
Internal business apps	24 hours

✓ **4 hours shows seriousness without overengineering**

RPO – Recovery Point Objective Explanation

The recommendation is ≤ 15 minutes

This mean:

- Maximum acceptable data loss: **15 minutes**
- Anything older must be recoverable

This matters because the platform supports:

- Submissions
- Uploaded evidence
- Certification actions

Even **small data loss** can:

- Create legal or audit issues
- Undermine trust

Comparison:

System Type	RPO
Financial ledgers	0–5 minutes
Regulatory record systems	5–30 minutes
Internal tools	Hours

✓ **15 minutes is conservative and defensible**

Sanity Check

Are these too strict? No.

These targets:

- Do **not** require:
 - Global multi-region active/active
 - 24/7 SRE teams
 - Extreme cost
- **Do** require:
 - Managed database backups
 - Health monitoring
 - Clear incident procedures

This is the **sweet spot** for a national-grade system.

About AI Usage

The PROD environment includes **AI-powered features** to enhance document understanding and user experience.

AI could be used for:

- AI-assisted reading and analysis of uploaded documents

- Retrieval-Augmented Generation (RAG) using indexed documents
- Summarization and explanation of reports and evidence files
- AI-supported user interactions within the application

AI usage in PROD is:

- User-triggered
- Interactive
- Integrated into core workflows

Unlike STAGING, PROD AI usage is expected to **grow gradually over time**, but remains bounded by application workflows (not open-ended chat usage).

For PROD sizing and cost modeling, the following **baseline assumptions** are used:

- **Initial phase:** ~2,000–3,000 AI requests per month (**b**roader user base, low adoption)
- **Mature phase:** 10,000+ AI requests per month (**N**ational rollout)

AI usage is expected to grow as features mature and user adoption increases.

Per-request token assumptions:

- **New input (non-cached):** ~5,000 tokens
- **Cached / repetitive input:** ~3,000 tokens
- **AI output:** ~1,500 tokens (1,500 tokens correspond to approximately 1,100–1,200 words, or 2–3 Word document pages, representing long-form AI responses)

These values are intentionally conservative and may increase as AI adoption grows.

The following are **not part of PROD AI usage (out of scope)**:

- High-volume conversational AI
- Large-scale document ingestion
- Batch or scheduled AI processing
- SLA-driven or latency-critical AI workloads
- Public or anonymous AI access

We suggest to use Azure Functions in PROD for:

- Orchestrating **LangChain workflows**
- Coordinating:
 - LLM calls
 - Retrieval (RAG) steps
 - Prompt assembly
 - Response post-processing
- Triggered **only by user actions**
- No background, batch, or scheduled AI jobs
- No public or anonymous access

All heavy computation (LLM inference, embeddings) is executed by **external managed AI services**, not inside the Function runtime.

Although the absolute number of AI requests in STAGING and early PROD may appear similar, the usage patterns differ significantly. STAGING AI usage is concentrated during demos, QA, and pilot scenarios with repeated flows, while PROD AI usage starts conservatively and grows gradually as adoption increases across organizations. For this reason, initial PROD sizing assumes modest AI usage, with clear headroom for future expansion.

About the Background Tasks

The analysis identifies three categories of background tasks needed for the app:

1 Database Cleanup

Examples include:

- Deleting obsolete or expired records
- Clearing orphaned data
- Running lightweight maintenance logic

Typical execution time: 20–60 seconds

Frequency assumption: 1 execution per hour

2 Reminders & Pending Tasks

Used to:

- Check for pending workflows
- Send notifications (email, webhook, queue messages)
- Trigger periodic application-level checks

Typical execution time: 1–10 seconds

Frequency assumption: Every 10 minutes

3 Data Lifecycle Management

Examples include:

- Archiving old documents
- Moving data between storage tiers
- Expiring or reclassifying items

Typical execution time: 5–30 seconds

Frequency assumption: Every 2 hours



Estimated Monthly Executions

Based on the execution patterns:

Task Type	Frequency	Estimated Monthly Executions
Database Cleanup	1/hour	~720/month
Reminders	1/10 minutes	~4,300/month
Data Lifecycle	1/2 hours	~360/month

Total estimated executions: ~5,400 executions/month

Execution Duration Assumptions

Typical durations are:

Task Type	Duration
Database Cleanup	~40 seconds
Reminders	~5 seconds
Data Lifecycle	~15 seconds

Average memory: **512 MB (0.5 GB)**

These durations fall well within serverless best practices and do not require premium or dedicated hosting.

Summary

The PROD environment is characterized by:

- Country-level usage
- Predictable, steady traffic
- Stateless backend architecture
- Relational transactional database
- Long-term document storage
- Production-grade observability and security
- AI features integrated into core workflows

These assumptions provide a **stable, scalable foundation** for capacity planning, cost estimation, and infrastructure design, while allowing controlled growth over time.